MILITARY SPECIFICATION

SIMULATOR, COMPRESSOR BLEED AIR SMU-65/E

1. SCOPE

1.1 This specification covers a pneumatic, bleed air compressor simulator, designated SMU-65/E.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-636	Box, Fiberboard
PPP-C-843	Cushioning Material, Cellulosic
PPP-C-1120	Cushioning Material, Uncompressed Bound Fiber for Packaging
PPP-T-60	Tape, Pressure Sensitive Adhesive, Waterproof, for Packaging

Military

MIL-P-116	Preservation, Methods of
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-W-16878	Wire, Electrical, Insulated, High Temperature
MIL-T-21200	Test Equipment for use with Electronic and Fire Control Systems, General Sepcification for
MIL-T-38191	Test Set, Pressure-Temperature TTU-205B/E

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:43Z
Check the source to verify that this is the current version before use.

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-470	Maintainability Program Requirements (for Systems and Equipments)
MIL-STD-471	Maintainability Demonstration
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods
MS3101	Connector, Plug, Electric, Cable Connecting, Solder Contacts, AN Type
MS3106	Connector, Plug, Electric, Straight, Solder Contacts, AN Type
MS3116	Connector, Plug, Electric, Solder Type, Straight, Bayonet Coupling
MS17322	Meter, Time Totalizing, Miniature, Digital, 115 Volt 400 Hertz
MS28741	Hose Assembly, Detachable End Fitting, Medium Pressure

Air Force - Navy Aeronautical

AN6270	Hose Assembly - Detachable Swivel Fitting, Low Pressure

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 General specification. The requirements of MIL-T-21200 as specified for class 2 equipment apply as requirements of this specification with the exceptions and additions specified herein. When the two specifications conflict, this specification shall govern.

3.3 Components. The simulator shall consist of a combination case; a control panel with all necessary switches, controls, displays, and connections; a pneumatic supply system in conjunction with a suitable pneumatic control system; an electrical power cable assembly; and pneumatic hose assemblies.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:43Z
Check the source to verify that this is the current version before use.

3.4 Design and construction. The simulator shall be designed for functional operation and use at both organizational (flight line) and intermediate (field shop) levels of maintenance. The simulator shall be used in conjunction with a TTU-205B/E test set conforming to MIL-T-38191, or equivalent, during ground level checkout of the central air data computer (CADC) in the F-4 aircraft, and shall perform the following primary functions:

a. Provide a continuous, regulated source of clean, dry, air pressure to simulate the engine compressor bleed air (P_a) required to operate a pneumatic valve in the static pressure compensator (SPC) module of the CADC

b. Provide simultaneously with the simulated engine bleed air, a continuous, regulated vacuum source to serve as a vacuum sink into which the exhaust air (P_e) of the SPC can flow

c. Perform the functions specified in a and b in such a manner that an accurate check of the SPC operation can be made at any simulated altitude from -1,000 to +40,000 feet.

3.4.1 General layout. The simulator shall be an integral unit, self-contained within its case. The case shall serve to house and protect the complete system and shall have a compartment in the cover for storage of all accessories. All interior components of the simulator shall be so mounted that the entire system can be completely exposed for maintenance purposes. Physical layout and operation of the indicators and controls shall be in accordance with basic human factors design criteria.

3.4.2 Configuration. The total volume of the case shall not exceed 1.75 cubic feet. The case shall be rectangular in shape with the control panel mounted horizontally to be viewed from above, or so inclined as to be viewed from the front of the case with the cover open. If mounted horizontally, the simulator shall be of such design as to meet the requirements specified herein when the control panel is in a vertical attitude. If inclined, the inclination shall be not less than 40° nor more than 50° from the horizontal. The external dimensions of the case, with case cover closed, shall not exceed 18 inches.

3.4.3 Maintainability. Maintainability of the simulator shall be considered of major importance. Maintainability effort shall be in effect throughout the design and manufacture of the simulator, and shall be accomplished in accordance with the design guidelines of MIL-STD-470, paragraph entitled Establish Maintainability Design Criteria. The guidelines shall be applied wherever applicab.., and wherever they do not conflict with other requirements specified herein. Maintenance tasks shall be performed at the lowest practicable maintenance level using, where practicable, only common handtools and equipment normally available for maintenance of similar equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:43Z
Check the source to verify that this is the current version before use.

3.4.3.1 Maintenance downtime. The simulator shall have a mean corrective maintenance downtime of 1 hour, and a maximum corrective maintenance downtime of 3 hours, as accomplished by 3 and 5 level repair technicians. Not more than two repair technicians shall be required simultaneously to perform any maintenance task. Periodic overhaul and complete calibration shall be considered as preventive maintenance, and shall not be included as corrective maintainance tasks.

3.4.4 Reliability. Reliability of the simulator shall be considered of prime importance throughout its design and manufacture. Reliability considerations shall inbiude, but not be limited to, standardization of parts, factors of safety by derating parts, functional application of parts, et cetera. The reliability concepts of MIL-STD-781 shall be adhered to during the design and manufacture of the simulator. The mean-time-between-failures (MTBF) shall be at least 500 hours at a 90-percent confidence factor.

3.4.4.1 Longevity. The simulator shall be capable of operating satisfactorily for 2,000 hours with reasonable servicing and maintenance and still meet all the performance requirements specified herein.

3.4.5 Control panel. The simulator shall incorporate a metal control panel upon which shall be mounted all necessary controls, displays, switches, lights, and external connectors required for compliance with the requirements of this specification.

3.4.6 Electrical overload protection. Circuit breakers shall be provided for protecting the pump and motor assembly. The overload protection features shall also protect the simulator should it be connected to a power source that would damage the unit.

3.5 Performance. The simulator shall be so designed and constructed that no degradation in performance below the requirements of this specification will result when the simulator is subjected to the following conditions:

3.5.1 Service conditions, electrical. The electrical power requirements of MIL-T-21200 as specified for class 2 equipment shall apply, except as specified herein.

3.5.1.1 Frequency. The simulator shall not be required to operate from a 60-Hz power supply.

Source: https://assist.dla.mil -- Downloaded: 2026l0-26T04:43Z
Check the source to verify that this is the currentversion before use.

3.5.2 Service conditions, environmental. The environmental requirements of MIL-T-21200 as specified for class 2 equipment shall apply, except as specified herein.

3.5.2.1 Altitude. The operating altitude range shall be -1,000 to +10,000 feet. The nonoperating altitude range shall be -1,000 to +50,000 feet.

3.5.2.2 Temperature. The operating temperature range shall be -30° to $+50^{\circ}$C.

3.5.2.2.1 Warmup. The maximum allowable warmup time at any temperature from -15° to $+50^{\circ}$C shall be 5 minutes. The maximum allowable warmup time at any temperature from -30° to -14°C shall be 10 minutes.

3.5.2.2.2 Stability of operation. After the allowable warmup period, the simulator shall maintain stable pressure control and regulation for at least 30 minutes at any temperature from -30° to $+50^{\circ}$C.

3.6 Pneumatic supply system. The pneumatic supply system required to accomplish the functions specified in 3.4 herein shall be provided by one or more electrically powered pneumatic pumps, which may be either wet or dry type. If wet-type pumps are used, each pump assembly shall be sealed to preclude loss of lubricant during use, and shall be designed for easy draining and filling without spillage of lubricant on the simulator. A change of lubricant shall not be required to meet the environmental requirements specified herein for both operating and nonoperating conditions. The pneumatic supply system shall not transmit sufficient vibration to the indicators to cause indicator movements greater than the indicator resolution requirements.

3.6.1 Pressure output. The simulator shall provide a regulated pressure output, to simulate engine compressor bleed air, and shall supply clean, dry air at minimum airflow rates of 0.15 pound per minute at 15 psig, 0.25 pound per minute at 30 psig, and 0.30 pound per minute at 50 psig under any operating conditions specified herein.

3.6.2 Vacuum source. The simulator shall provide a regulated vacuum source into which the SPC exhaust air can flow. Except as specified in 3.6.2.1, the simulator, when used under any operational conditions as specified herein, shall be capable of maintaining the vacuum source within ±½ inch Hg of the pressure equivalent to the altitude being simulated by the ITU-205B/E test set, or equal. The simulator shall stabilize at the required vacuum within 1 minute after the test set has stabilized at any selected altitude from -1,000 to +40,000 feet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:43Z
Check the source to verify that this is the current version before use.

3.6.2.1 CADC safety relief valve. Under any operational or environmental condition specified herein, the simulator shall not cause actuation of the CADC safety relief valve. This shall be accomplished within the simulator by assuring that the pressure of the vacuum source (P_{vs}) can never exceed ambient atmospheric pressure (P_a). This condition can only occur when the indicated static pressure (P_s) being provided by the TTU-205B/E, or equivalent, test set exceeds ambient atmospheric pressure (ie, when the TTU-205B/E or equivalent test set is simulating an altitude below ambient). Whenever P_s exceeds Pa, Pvs shall be referenced to P_a, and P_{vs} may, therefore, differ from P_s by more than ±½ inch Hg. The values and tolerances of P_{vs}, in inches Hg, shall be as follows:

$$P_{vs} = P_s \pm 1/2, \text{ except when } P_s > P_a$$

$$\text{when } P_s > P_a, \; P_{vs} = P_a$$

3.7 Pressure control system. The control of the generated pressure shall be accomplished by a single, multiposition pressure regulator. The regulator shall be capable of controlling the pressure output of the simulator at any selected pressure from 14 psig to the maximum capability of the pump. Accuracy of the pressure control shall be within ±1 psi.

3.8 Functional displays. The functional displays described herein shall be designed to provide maximum visibility to the operator, and shall be easily replaceable at field level.

3.8.1 Pressure indicator. A pressure indicator, calibrated directly in terms of psi, shall be provided on the control panel to present the regulated pressure output of the simulator to within an accuracy of at least ±1 psi. The indicator shall be clearly readable to at least 1 psi, and shall have sufficient range to indicate any pressure from 14 to 60 psig.

3.8.2 Null indicating meter. A zero-center null indicating meter shall be provided on the control panel to indicate stabilization of the vacuum source. The meter shall indicate whether or not the vacuum source is within the tolerance specified in 3.6.2.

3.9 Pressure limit safety device. Positive means shall be incorporated into the simulator control system to prevent the pressure being provided to the system under test from exceeding 60 psig.

3.10 Accessories. The following accessories shall be provided as part of the simulator, and shall be capable of withstanding the environmental conditions specified herein. The accessories shall not be permanently attached to the simulator.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:43Z
Check the source to verify that this is the current version before use.

3.10.1 Power cable assemblies. Two separate power cable assemblies shall be provided for connecting the simulator to an external power source. The cable assemblies shall be as specified in MIL-T-21200, except as follows:

3.10.1.1 Simulator to test bench. One cable assembly shall be 8 feet in length and shall permit direct interconnection between the simulator and the F-4 aircraft CADC test bench. This cable shall be terminated with an MS3106R18-5S, or equivalent, connector and an MS3106E32-17P, or equivalent, connector providing interconnection between the simulator and test bench, respectively.

3.10.1.2 Simulator to aircraft. The remaining cable assembly shall be 12 feet in length and, when used in conjunction with the cable assembly specified in 3.10.1.1, shall permit direct interconnection between the simulator and the AN/AWA-6A cooling pumping cart junction box for flight line use. This cable shall be terminated with an MS3101E-32-17S, or equivalent, connector and an MS3116F-12-3P, or equivalent, connector providing interconnection between the cable specified in 3.10.1.1 and the AN/AWA-6A cooling pumping cart junction box, respectively.

3.10.2 Pneumatic hose assemblies. Pneumatic hose assemblies conforming to N6270 and MS28741 shall be provided as follows:

a. One AN6270-6 hose assembly for the Pe interconnection between the simulator and the aircraft. The aircraft end of the assembly shall terminate with a No. 4 quick-disconnect coupling and an adapter to convert from AN6270-6 hose to the No. 4 coupling, providing interconnection capability to all F-4 series aircraft.

b. One MS28741-5 hose assembly for the Pr interconnection between the simulator and the aircraft. The aircraft end of the assembly shall have a No. 5 quick-disconnect coupling to provide interconnection with all F-4 series aricraft.

c. One AN6270-4 and AN6270-6 combination hose assembly for Psi interconnection between the simulator, the pneumatic test set, and the aircraft. The simulator end of the assembly shall have an AN6270-4 hose fitting, the pneumatic test set end of the assembly shall have an AN6270-6 hose fitting, and the aircraft end of the assembly shall have a No. 6 quick-disconnect coupling to provide interconnection with all F-4 series aircraft.

3.11 Elapsed time indicator. A 10,000-hour elapsed time indicator conforming to MS17322-2 shall be provided to indicate pump running time. The indicator mechanism shall be so mounted on the control panel that the elapsed time indicator display will be readily visible to the operator.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:43Z
Check the source to verify that this is the current version before use.

3.12 Wire. Unless otherwise specified (see 6.2), all wire shall be in accordance with MIL-W-16878, type C, D, or E.

3.13 Weight. The weight of the completely assembled simulator, including the case and accessories, shall not exceed 65 pounds.

3.14 Operating instructions. A complete set of operating instructions shall be provided with the simulator and shall be stored in the case cover. The instructions shall be printed on material of durable nature, and shall withstand the environmental conditions specified herein without noticeable deterioration. The instructions shall concisely outline the procedures to be followed in using the simulator.

3.15 Diagrams. Wiring and piping diagrams shall be mounted on the inside of the simulator in a location easily accessible for ready reference during servicing operations. The diagrams shall be permanent in nature and as large as practicable. The diagrams shall contain pertinent information about the electrical and pneumatic systems of the simulator and shall withstand the environmental tests specified herein without noticeable deterioration.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the simulator shall be in accordance with MIL-I-21200 with the additions specified herein.

4.3 Test conditions

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:43Z
Check the source to verify that this is the current version before use.

4.3.2 Test standard. The accuracy of the pressure indicator and the ability of the pump to provide the specified rates of flow shall be determined by comparison with a pressure standard and through the use of an accurately calibrated flowmeter, respectively.

4.3.3 Attitude. Unless otherwise specified, the simulator shall be tested in the normal operating position.

4.3.4 Line voltage. Unless otherwise specified, the simulator shall be supplied with 115V ac, single-phase, 400-Hz power. The voltage and frequency shall be maintained within ±2 percent.

4.3.5 Allowable warmup time. Prior to performing any inspection or measurement, the allowable warmup time shall not exceed the limits specified in 3.5.2.2.1 herein.

4.3.6 Accessory testing. One of each accessory; including cable assemblies, operating instructions, schematic diagrams, et cetera, shall be subjected along with the simulator to the environmental tests described under 4.6.5 herein.

4.4 Preproduction testing (see 6.2)

4.4.1 Test samples. The test samples shall consist of six simulators representative of the production equipment.

4.4.2 Preproduction tests. The preproduction tests shall consist of all tests described under 4.6.

4.4.2.1 Preproduction test distribution. The distribution of the preproduction tests on the six test samples shall be as follows:

a. Three of the test samples shall be subjected to all tests described under 4.6 herein, except 4.6.6 (reliability). Each test sample need not be subjected to all tests. The tests shall be distributed among the three samples so as to minimize the real time and the cost required to perform the tests.

b. The remaining three samples shall be subjected to the reliability tests described in 4.6.6.

4.5 Acceptance tests. The acceptance tests shall consist of the following:

a. Individual tests . . . . . See 4.5.1

b. Sampling plans and tests. . . . . See 4.5.2

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:43Z
Check the source to verify that this is the current version before use.

4.5.1 Individual tests. Each simulator shall be subjected to the following tests as described under 4.6:

a. Examination of product 4.6.1

b. Functional tests 4.6.2

c. Position error 4.6.2.4.1

4.5.2 Sampling plans and tests. Sampling tests shall not be performed unless specified by the procuring activity (see 6.2). When required and not otherwise described, the sampling tests shall be performed as specified herein.

4.5.2.1 Sampling plan A. Three simulators shall be selected at random from each 100 or less produced on the contract or order and subjected to the following tests as specified under 4.6:

a. Individual tests

b. Low temperature test

c. High temperature test.

4.5.2.2 Sampling plan B. One simulator shall be selected at random from the first 15 produced on the contract or order and subjected to the following tests as described under 4.6:

a. Sampling plan A tests

b. Power variation test

c. Electromagnetic interference test

d. Low pressure test

e. Rain test

f. Humidity test

g. Dust test

h. Explosion test

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:43Z
Check the source to verify that this is the current version before use.

i. Vibration test

j. Shock test

k. Salt-fog test

l. Fungus test.

4.5.2.3 Sampling plan C. A minimum of three simulators shall be selected at random from the first 15 produced on the contract or order and subjected to the tests specified in 4.6.6 and 4.6.6.1.

4.5.2.4 Rejection and retest. When one item selected from a production run fails to meet the specification, items still on hand or later produced shall not be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.5.2.4.1 Individual tests may continue. For operational and production reasons, individual tests may be continued pending investigation of a sampling test failure; but final acceptance of items on hand or later produced shall not be made until it is determined that items meet all requirements of this specification.

4.5.3 Defects in items already accepted. Investigation of a test failure could indicate that defects exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods

4.6.1 Examination of product. The simulator shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, marking, dimensions, weight, ease of adjustment, and completeness of assembly.

4.6.2 Functional tests. Sufficient tests shall be performed to verify that the simulator can perform the functions specified in 3.4, and to verify that all components and accessories are properly assembled. The simulator, via the cable and hose assemblies specified in 3.10, shall be connected into

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:43Z
Check the source to verify that this is the current version before use.

a testing setup to simulate actual operational conditions and to permit verification of the performance capabilities of the simulator. As a minimum, the following tests shall be performed:

4.6.2.1 Pressure output tests. The output pressure and flow rates shall be tested as follows:

4.6.2.1.1 Accuracy. The pressure output of the simulator shall be connected to an accurately calibrated pressure standard. The simulator pressure regulation control shall be sequentially adjusted so that the pressure standard indicates 5 psig increments between the limits of 15 and 50 psig. While at each incremental pressure setting, a comparison shall be made to insure that the simulator output gage indications are within the specified tolerances. This test shall be performed as necessary to verify accuracy at the simulator output gage.

4.6.2.1.2 Pressure flow. The pressure output of the simulator shall be connected to an accurately calibrated flowmeter. The simulator pressure regulation control shall be sequentially adjusted so that the simulator output gage indicates pressures of 50, 30, and 15 psig. The flowmeter shall indicate minimum flow rates of 0.30, 0.25, and 0.15 pounds per minute, respectively. After each pressure setting, the pressure generator shall continue to operate for 30 minutes, remaining within 1 psig of the original setting without further adjustment. Measurements shall be made at the beginning and end of the 30-minute period required for each pressure. During this test, the flowmeter shall be exhausted into the same ambient conditions under which the simulator pump is operating.

4.6.2.2 Vacuum source test. The capability of the simulator to maintain a vacuum source as required by 3.6.2 shall be verified. The test shall verify that the vacuum source is maintained within the required tolerance under the following three simultaneous conditions:

a. While the simulator is providing all required pressures at the required flow rate of the SPC

b. While the SPC is exhausting into the simulator vacuum source

c. While the SPC is being supplied with pressures from a TTU-205B/E, or equivalent, test set to simulate altitudes from -1,000 to 40,000 feet, in increments no greater than 5,000 feet.

4.6.2.3 Pressure limit safety device test. The pressure limit safety device shall be operated to determine compliance with 3.9 herein.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:43Z
Check the source to verify that this is the current version before use.

4.6.2.4 Position error test. The simulator shall be tested to insure proper operation while in a vertical attitude (see 3.4.2). While the simulator is in the normal viewing position, proper operation shall be determined in accordance with the requirements of 4.6.2 herein. After proper operation has been established, the simulator shall be rotated 90° so that the front panel lies in a vertical plane. While in this attitude, simulator performance shall be identical to its horizontal attitude performance and shall be demonstrated by performance of the tests described in 4.6.2. During the demonstration, the stability test requirements may be satisfied with only the performance of the 15 psig sequence.

4.6.2.4.1 For acceptance testing, during the pressure flow test of 4.6.2.1.1, the simulator shall be tested to insure proper operation while in a vertical attitude. During the 50-psig phase of the pressure flow test, the simulator shall be rotated 90° so that the front panel lies in a vertical plane. While in this attitude, simulator performance shall be identical to its horizontal attitude performance. Measurements shall be made in both the horizontal and vertical positions.

4.6.3 Power variation test. The performance of the simulator shall be as specified herein, and shall be demonstrated by performing the functional tests specified in 4.6.2.1 through 4.6.2.3 while the simulator is being supplied with the following power variations:

a. 103.5V, 380 Hz

b. 103.5V, 420 Hz

c. 125.5V, 380 Hz

d. 126.5V, 420 Hz

4.6.4 Electromagnetic interference test. The simulator shall be subjected to electromagnetic interference in accordance with MIL-I-6181 as specified for class II (flight line) equipment.

4.6.5 Environmental tests. The following tests shall be performed to verify that the simulator can function satisfactorily under the environmental conditions specified herein:

4.6.5.1 Low temperature test. The simulator shall be subjected to low temperature in accordance with MIL-STD-810, method 502, procedure I. The performance of the simulator during this test shall be as specified herein, and shall be demonstrated at -30°C by performing the functional tests specified in 4.6.2. The performance of the simulator after this test shall be as specified herein, and shall be demonstrated under standard test conditions by performing the functional tests specified in 4.6.2.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:43Z
Check the source to verify that this is the current version before use.

4.6.5.2 High temperature test. The simulator shall be subjected to high temperature in accordance with MIL-STD-810, method 501, procedure I, except the maximum nonoperating temperature shall be 85°C. The performance of the simulator during this test shall be as specified herein, and shall be demonstrated at 50°C by performing the functional tests specified in 4.6.2. The performance of the simulator after this test shall be as specified herein, and shall be demonstrated under standard test conditions by performing the functional tests specified in 4.6.2.

4.6.5.3 Low pressure test. The simulator shall be subjected to low pressure in accordance with MIL-STD-810, method 500, procedure I. The performance of the simulator during this test shall be as specified herein, and shall be demonstrated by performing the functional tests specified in 4.6.2. In addition, compliance with 3.6.2.1 to prevent actuation of the CADC safety relief valve shall be demonstrated during this test. The performance of the simulator after this test shall be as specified herein, and shall be demonstrated under standard test conditions by performing the functional tests specified in 4.6.2.

4.6.5.4 Rain test. With the case cover removed, the simulator shall be subjected to rain in accordance with MIL-STD-810, method 506, procedure I. The performance of the simulator after this test shall be as specified herein, and shall be demonstrated under standard test conditions by performing the functional tests specified in 4.6.2.

4.6.5.5 Humidity test. With the case cover removed, the simulator shall be subjected to humidity in accordance with MIL-STD-810, method 507, procedure I. The performance of the simulator after this test shall be as specified herein, and shall be demonstrated under standard test conditions by performing the functional tests specified in 4.6.2.

4.6.5.6 Dust test. With the case cover removed, the simulator shall be subjected to dust in accordance with MIL-STD-810, method 510, procedure I. The performance of the simulator after this test shall be as specified herein, and shall be demonstrated under standard test conditions by performing the functional tests specified in 4.6.2.

4.6.5.7 Explosion test. With the case cover removed, the simulator shall be subjected to explosion in accordance with MIL-STD-810, method 511, procedure I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:43Z
Check the source to verify that this is the current version before use.

4.6.5.8 Vibration tests. The simulator shall be subjected to vibration tests as follows:

4.6.5.8.1 Test number 1. While in a nonoperating condition, the simulator shall be vibrated with the frequency varying logarithmically from 5 to 55 Hz, and returning to 5 Hz, in approximately 7.5 minutes. The simulator shall be observed for resonant frequencies throughout the vibration cycle. Three complete test cycles (5-55-5) shall be performed along each of three mutually perpendicular axes, for a total of 22.5 minutes of vibration along each axis. The double amplitude (peak-to-peak) of the vibration shall be as follows:

Frequency Range (Hz)	Double Amplitude (Inch)
5 to 15	0.06
16 to 25	0.04
26 to 55	0.02.

4.6.5.8.2 Test number 2. The simulator shall be vibrated for a period of 30 minutes at each of the four most severe resonant frequencies observed in test number 1. The vibration shall be at the same amplitude, and along the same axis, at which the most severe resonant reaction was observed.

4.6.5.8.3 Alternate A to test number 2. If less than four (but at least one) resonant frequencies are observed in test number 1, the simulator shall be vibrated for 120 minutes under the conditions of test number 2, except the length of time for vibration at each resonant frequency shall be determined by equally distributing the total test time (120 minutes) among the observed resonant frequencies.

4.6.5.8.4 Alternate B to test number 2. If no resonant frequencies are observed in test number 1, the simulator shall be vibrated with the frequency varying logarithmically from 5 to 55 Hz and returning to 5 Hz in approximately 7.5 minutes. Eight complete test cycles (5-55-5) shall be performed along each of three mutually perpendicular axes, for a total of 60 minutes along each axis. The double amplitude of the vibration shall be as specified for test number 1.

4.6.5.8.5 Post test evaluation. The performance of the simulator after the vibration test shall be as specified herein, and shall be demonstrated under standard test conditions immediately following test number 2 (or the applicable alternate test) by performing the functional tests specified in 4.6.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:43Z
Check the source to verify that this is the current version before use.

4.6.5.9 Shock test. The simulator shall be subjected to shock in accordance with MIL-STD-810, method 516, procedure I. The performance of the simulator after this test shall be as specified herein, and shall be demonstrated under standard test conditions by performing the functional tests specified in 4.6.2.

4.6.5.10 Salt-fog test. With the case cover removed, the simulator shall be subjected to salt-fog in accordance with MIL-STD-810, method 509, procedure I. The performance of the simulator after this test shall be as specified herein, and shall be demonstrated under standard test conditions by performing the functional tests in 4.6.2.

4.6.5.11 Fungus test. If the simulator is constructed entirely of fungus inert materials, a certified statement may be provided verifying that no fungus nutrient materials have been used in the construction of the item. If requested by the procuring activity, compliance shall be demonstrated by submitting a complete list of parts and materials. If fungus nutrient materials are used, fungus testing shall be required, and shall be performed, with the case cover removed, in accordance with MIL-STD-810, method 508, procedure I.

4.6.6 Reliability test. A minimum of three simulators shall be subjected to a reliability test under actual electrical operation for a minimum total of 1,152.5 hours. This test shall be conducted in accordance with environmental test level E of MIL-STD-781, except the temperature cycling limits shall be -30° and $+50^{\circ}$C. Performance checks shall be made at intervals not exceeding 100 hours, and shall include the tests specified in 4.6.2. The simulators shall be tested until an accept or reject decision has been reached. Electrical operation shall be at 50 psig and 0.35 pound per minute. The cycle specified herein shall be repeated until termination of this test.

4.6.6.1 Reliability accept-reject criteria. Completion of 1,152.5 hours of operation with no failures will permit acceptance. Failure is considered as any condition that results in out-of-tolerance performance. If failures occur, testing shall be continued as specfied in 4.6.6 until acceptance or rejection is determined in accordance with table I. Compliance with table I will demonstrate a minimum MTBF of 500 hours at a 90-percent confidence level. No equipment from a production run of which the test group is a part shall be shipped until an accept decision has been reached on that test group.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:43Z
Check the source to verify that this is the current version before use.

TABLE I

Accept-Reject Criteria

Failures	Total Test Time (Hours
0	1,152.5
1	1,945
2	2,650
3	3,350
4	4,000
5	4,625
Over 5	Equipment will be rejected subject to redesign as required to demonstrate the specified MTBF.

4.6.6.2 Longevity. After satisfactory completion of the reliability test, two simulators shall continue to be cycled under the reliability test conditions until a total of 2,000 hours operating time has accumulated on each simulator. All failures which occur during this testing shall be recorded and analyzed. Performance during and after this test shall be verified by the same test procedures used for the reliability test, except that performance checks shall be made at 200-hour intervals. The operating time required to perform the performance checks, as recorded on the elapsed time indicator, shall be counted as part of the 2,000-hour operating requirement.

4.6.6.2.1 Longevity accept-reject criteria. Failures due to component wearout shall be cause for rejection. A wearout failure shall be any predictable, time-dependent failure which cannot be repaired by calibration or adjustment.

4.6.6.2.2 Replacement of parts. Replacement of major components or assemblies shall not be allowed during the reliability and longevity tests. The same major components and assemblies which are in the simulator at the beginning of the reliability test shall be used in the simulator throughout the reliability and longevity tests. Replacement of small detail parts, such as resistors, capacitors, and valve seals, will be allowed if the replacement has been approved by the procuring activity prior to the start of the reliability test, or if the part randomly fails during test. If a failure occurs which necessitates repair of a major component or assembly, the test shall be discontinued (on the failed simulator), and the component or assembly repaired. The same component or assembly shall then be returned

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:43Z
Check the source to verify that this is the current version before use.

to the simulator and testing resumed. If a nonrepairable failure occurs to a major component or assembly, the test shall be discontinued and the procuring activity shall be given complete details of the failure. Testing shall not be resumed unless authorized by the procuring activity.

4.6.7 Maintainability demonstration. Compliance with the maintainability requirements shall be demonstrated by a maintenance demonstration procedure, maintenance task selection, and maintenance task performance in accordance with MIL-STD-471. Maintenance task selection shall be in accordance with appendix A, and the tasks shall be performed in accordance with appendix B, method 1, test plan pair A_1 plus B_1. Performance of the tasks shall be observed by appropriately trained personnel who shall record maintenance downtimes, tools and equipment required, obstructions or conditions hindering the tasks, and potential hazards to personnel. Depot level maintenance tasks need not be performed, but shall be delineated with a prediction of the downtime and tools required for each task. Signs of excessive wear or other indications of potential simulator failure shall be recorded when observed and the information supplied to the simulator design group for study and recommendations. Such recommendations shall be subject to approval by the procuring activity before being implemented.

4.7 Inspection of preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to determine conformance to the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A

5.1.1.1 Cleaning and drying. The simulator shall be cleaned in accordance with process C-1 and dried in accordance with procedures D-1 and D-4 of MIL-P-116.

5.1.1.2 Unit packaging. Each simulator shall be preserved and packaged in accordance with method IIB of MIL-P-116. The case shall serve as the interior container. Dessicant shall be placed inside the case with greaseproofed material conforming to MIL-B-121, type I, grade A, class 2, between the dessicant and bare metal. Handles, corners, and other protrusions shall be padded with cushioning material conforming to PPP-C-843 secured in place with tape conforming to PPP-T-60. The exterior container of the method IIB pack shall be as specified for the applicable level of pack (see 5.2.1 or 5.2.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:43Z
Check the source to verify that this is the current version before use.

5.1.2 Level C. Each simulator shall be preserved and packaged to afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity. This level may conform to the supplier's commerical practice when it meets the requirements of this level.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. One simulator, preserved and packaged as specified in 5.1.1, shall be packed in an exterior container conforming to PPP-B-636, variety double-wall, class weather-resistant. The exterior container shall be lined with a minimum of 1 inch thickness of cushioning material conforming to PPP-C-1120. Containers shall be of uniform size and shape and of minimum tare and cube. Closure and strapping shall be in accordance with the appendix of the specification.

5.2.2 Level B. One simulator, preserved and packaged as specified in 5.1.1, shall be packed in a shipping container conforming to PPP-B-636, variety double-wall, class domestic.

5.2.3 Level C. The simulator, preserved and packaged as specified in 5.1.2, shall be packed to afford adequate protection against damage during shipment from the supply source to the first receiving activity. Shipping containers shall conform to the Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation. This may conform to the supplier's commercial practice when it meets the requirements of this level.

5.3 Physical protection. Interior cushioning, blocking, and bracing shall be in accordance with MIL-STD-1186.

5.4 Marking. In addition to any special marking required by the contract or order (see 6.2), interior packages and exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be:

SIMULATOR, COMPRESSOR BLEED AIR SMU-65/E

6. NOTES

6.1 Intended use. The SMU-65/E simulator is intended for organizational (flight line) and intermediate (field shop) use in testing the central air data computer and associated equipment used on the F-4 aircraft.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:43Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. When wire is to be other than as specified in 3.12

c. Location and conditions for preproduction testing (see 4.4)

d. When sampling plans A, B, and C are required (see 4.5.2)

e. Applicable levels of preservation, packaging, and packing required (see 5.1 and 5.2)

f. Special shipment marking (see 5.4).

Custodian:
Air Force - 11

Reviewer:
Air Force - 82

Preparing activity:
Air Force - 11

Project No. 4920-F282

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:43Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 68 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

AFLC-WPAFB-OCT 67 2M

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:43Z
Check the source to verify that this is the current version before use.

ASD/ENZS
Wright-Patterson AFB, O 45433

POSTAGE AND FEES PAID

UNITED STATES AIR FORCE
OFFICIAL BUSINESS

ASD/ENZS
Wright-Patterson AFB, O 45433

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:43Z
Check the source to verify that this is the current version before use.